

Table 2: IPCop interfaces

| Interface colour | Trust level                   | Typical function                                                                |
|------------------|-------------------------------|---------------------------------------------------------------------------------|
| Green            | 1 – The most trusted          | The internal network (or protected network)                                     |
| Blue             | 2 – Semi-trusted              | Wireless network in separate subnet                                             |
| Orange           | 3 – DMZ (De-militarised zone) | For resources being accessed from internal systems as well as from the Internet |
| Red              | 4 – Non-trusted               | The Internet connection                                                         |

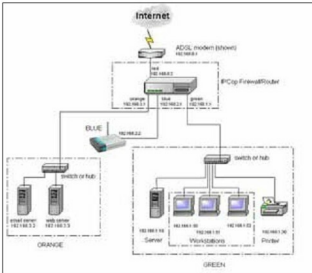


Diagram 1: IPCop network

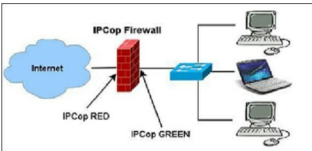


Diagram 2: IPCop Red – Green

One can fine-tune the IPCop installation depending on organisational requirements. For example, only a Green + Red network is sufficient for a basic office firewall. A Green + Red + Blue network will suit an office network with wireless access or two separate subnets. If a few servers are required to be accessed from the internal network as well as the Internet, a DMZ can be added.

During the installation process, IPCop probes, detects and allots various network cards to the corresponding interfaces. Typically, the on-board Ethernet is detected first, a card in the first PCI slot is detected second, and so on. Regarding a Green + Red network, by default, all traffic from the Internal network (Green) to the Internet (Red) is allowed, whereas all traffic from Red to Green is blocked. An excellent self-explanatory network diagram of a typical IPCop network is available at

<http://ipcop.org/1.4.0/en/install/html/decide-configuration.html>, which is reproduced here for ready reference.

## Installation

Download the IPCop 1.4.20 stable release, and prepare a boot CD. Don't forget to take a back-up of the intended install system if necessary—all the partitions and data on it will be wiped out. Boot from the IDE optical drive on the firewall system. SATA drives are not supported by default—you may be able to use BIOS settings on some motherboards to map a SATA optical drive as IDE. IPCop installation is straightforward and simple. Table 3 has the steps to install a Green + Red network.

Table 3: IPCop installation steps

| Step                               | Selection and explanation                                                                             |
|------------------------------------|-------------------------------------------------------------------------------------------------------|
| Choose language                    | English                                                                                               |
| Welcome screen                     | Continue                                                                                              |
| Selection of boot device           | CD / USB key / HTTP / FTP server - Select CD-ROM                                                      |
| Whether to restore from old backup | No back-up, skip the step                                                                             |
| Detection of Ethernet card         | Probe                                                                                                 |
| Green IP address                   | Enter the correct IP address (IP used for test set-up: 192.168.101.1)                                 |
| Reboot screen                      | First installation step OK. Partitions created, IPCop installed                                       |
| Keyboard mapping                   | Default                                                                                               |
| Time-zone                          | Select Asia / Kolkata                                                                                 |
| Hostname                           | IPCop (You may enter the desired hostname)                                                            |
| Domain                             | Firewall (You may enter the desired hostname)                                                         |
| ISDN set-up                        | Disable ISDN, we shall use static IP instead                                                          |
| Network configuration type         | Green + Red                                                                                           |
| Drivers and card assignment        | Probe for additional cards, assign the detected card to the Red interface                             |
| Address settings                   | PPPoE                                                                                                 |
| DNS and gateway                    | Applicable only for a static IP address                                                               |
| Passwords                          | Allot root (Shell access), admin (Web GUI access) and back-up (for restoration of back-ups) passwords |